ParcelFlow Regression Test Case Catalogue

SYNTHETIC DEMO DATA - ParcelFlow is a fictitious demo application. Confidential - DXC internal use for the Converge Intelligent Test Suite Selection demo.

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Requirements

Requirement	Statement	Notes
REQ-QUOTE-1	POST /api/quotes accepts integer weight_g (1-50000), enum destination, enum service.	No floats for weight or money.
REQ-QUOTE-2	Quote amount is computed in integer cents from a versioned rate card.	reference-data/rate_card.json
REQ-QUOTE-3	EXPRESS service is only quotable when feature_flags.express_service_enabled is true for the active environment.	Baseline: disabled in ST/SIT/UAT.
REQ-AUTH-1	Every quote/booking request must present a known role (customer/ops/admin) via X-Demo-Role, checked against the environment's allow-list.	Demo-only; not production auth.
REQ-BOOK-1	Booking confirmation retries against the simulated carrier gateway up to gateway.retry_count additional times on timeout.	decline is never retried.
REQ-BOOK-2	If retries are exhausted and gateway.fallback_enabled is true, the booking is queued via fallback rather than failing.	
REQ-VALID-1	Request bodies larger than 16KB are rejected before JSON parsing.	validation.MAX_BODY_BYTES
REQ-VALID-2	Unknown fields in request bodies are rejected.	Strict shape validation.
REQ-CONFIG-1	Configuration documents are validated for known environment name, known role names, and non-negative integer gateway timeout/retry values.	config.py
REQ-SEC-1	Secret reference values must never appear in error messages or logs.	Enforced by config.py validation errors and app.py error handlers.

Test Case Catalogue

Rec	Test ID	Title	Type	Description	Selector / Manual steps	Components	Reqs	Cycles	Crit.	Packs	Deps	Dur (s)
1/31	TC-001	Health endpoint reports environment	automated	GET /health returns 200 with the active environment name.	python -m unittest tests.test_api.HealthEndpointTests.test_health_returns_environment	src/parcelflow/app.py	-	ST, SIT, UAT, REGRESSION	low	smoke-pack	-	1
2/31	TC-002	Domestic standard quote uses integer-cents arithmetic	automated	compute_quote_cents(weight_g=500, DOMESTIC, STANDARD) returns an exact int equal to base_cents + per_gram_cents*weight_g, never a float.	python -m unittest tests.test_pricing.PricingTests.test_domestic_standard_quote_is_integer_cents	src/parcelflow/pricing.py reference-data/rate_card.json	REQ-QUOTE-2	ST, SIT, REGRESSION	high	financial-accuracy-pack	-	1
3/31	TC-003	Quote weight ceiling is enforced	automated	compute_quote_cents rejects weight_g above MAX_WEIGHT_G with InvalidQuoteRequest.	python -m unittest tests.test_pricing.PricingTests.test_weight_exceeding_maximum_is_rejected	src/parcelflow/pricing.py src/parcelflow/models.py	REQ-QUOTE-1	ST, SIT, REGRESSION	medium	financial-accuracy-pack	-	1
4/31	TC-004	EXPRESS service respects the express feature flag	automated	compute_quote_cents rejects service=EXPRESS when express_enabled is False.	python -m unittest tests.test_pricing.PricingTests.test_rejects_express_when_feature_flag_disabled	src/parcelflow/pricing.py config/st.json config/sit.json config/uat.json	REQ-QUOTE-3	ST, SIT, UAT, REGRESSION	high	-	-	1
5/31	TC-005	Config loader rejects unknown environment names	automated	load_config_file raises ConfigValidationError when environment is not in {ST, SIT, UAT}.	python -m unittest tests.test_config.ConfigLoadingTests.test_rejects_unknown_environment_name	src/parcelflow/config.py	REQ-CONFIG-1	ST, SIT, REGRESSION	medium	-	-	1
6/31	TC-006	Config validation errors never echo secret reference values	automated	When a secrets.* value is present alongside an unrelated validation failure, the raised error message string does not contain that secret reference.	python -m unittest tests.test_config.ConfigLoadingTests.test_error_message_never_includes_secret_reference_value	src/parcelflow/config.py	REQ-SEC-1	ST, SIT, UAT, REGRESSION	high	security-pack	-	1
7/31	TC-007	Config loader accepts a well-formed environment document	automated	load_config_file parses a valid ST config and exposes gateway_retry_count and express_service_enabled correctly.	python -m unittest tests.test_config.ConfigLoadingTests.test_loads_valid_config_for_known_environment	src/parcelflow/config.py	REQ-CONFIG-1	ST, SIT, REGRESSION	medium	-	-	1
8/31	TC-008	Role not in allow-list is rejected	automated	auth.require_role raises AuthorizationError when role is not a member of allowed_roles.	python -m unittest tests.test_auth.AuthTests.test_disallowed_role_raises	src/parcelflow/auth.py	REQ-AUTH-1	ST, SIT, UAT, REGRESSION	high	security-pack	-	1
9/31	TC-009	Gateway retries once on timeout before succeeding	automated	GatewaySimulator with outcomes=[timeout, success] confirms on attempt 2 and reports attempts=2.	python -m unittest tests.test_gateway.GatewaySimulatorTests.test_timeout_then_success_retries_and_reports_attempts	src/parcelflow/gateway.py	REQ-BOOK-1	ST, SIT, REGRESSION	high	-	-	1

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10/31	TC-010	Decline outcome is never retried	automated	GatewaySimulator with outcomes=[decline, success] raises GatewayUnavailableError after exactly one attempt, even with retries remaining.	python -m unittest tests.test_gateway.GatewaySimulatorTests.test_decline_is_not_retried_even_with_retries_remaining	src/parcelflow/gateway.py	REQ-BOOK-1	ST, SIT, REGRESSION	high	-	-	1
11/31	TC-011	Fallback is used once retries are exhausted	automated	GatewaySimulator with retry_count=1, fallback_enabled=True, outcomes=[timeout, timeout] returns a FALLBACK-QUEUED confirmation with used_fallback=True.	python -m unittest tests.test_gateway.GatewaySimulatorTests.test_fallback_used_after_retries_exhausted_when_enabled	src/parcelflow/gateway.py	REQ-BOOK-2	ST, SIT, REGRESSION	high	-	-	1
12/31	TC-012	Booking persists and round-trips through SQLite	automated	insert_booking followed by get_booking returns the same quote_id and status that were inserted.	python -m unittest tests.test_db.DbTests.test_insert_and_get_booking_roundtrip	src/parcelflow/db.py	REQ-BOOK-1	ST, SIT, REGRESSION	medium	-	-	1
13/31	TC-013	Oversized request bodies are rejected before JSON parsing	automated	parse_quote_request raises InvalidQuoteRequest when body exceeds MAX_BODY_BYTES.	python -m unittest tests.test_validation.RequestValidationTests.test_parse_quote_request_rejects_oversized_body	src/parcelflow/validation.py	REQ-VALID-1	ST, SIT, UAT, REGRESSION	medium	security-pack	-	1
14/31	TC-014	Unknown fields in quote requests are rejected	automated	parse_quote_request raises InvalidQuoteRequest when the request body contains a field outside the fixed schema.	python -m unittest tests.test_validation.RequestValidationTests.test_parse_quote_request_rejects_unknown_field	src/parcelflow/validation.py	REQ-VALID-2	ST, SIT, REGRESSION	low	-	-	1
15/31	TC-015	End-to-end shipping quote and checkout verification	automated	Narrow assertion only: POST /api/quotes for a DOMESTIC/STANDARD/500g request returns amount_cents == 1000 as an int. Despite the title, this does not exercise checkout, booking, or any non-DOMESTIC/STANDARD path.	python -m unittest tests.test_api.QuoteEndpointTests.test_valid_quote_returns_integer_amount	src/parcelflow/app.py src/parcelflow/pricing.py	REQ-QUOTE-1, REQ-QUOTE-2	ST, SIT, REGRESSION	high	financial-accuracy-pack	-	2
16/31	TC-016	Quote request without a role header is forbidden	automated	POST /api/quotes without X-Demo-Role returns HTTP 403.	python -m unittest tests.test_api.QuoteEndpointTests.test_quote_without_role_header_is_forbidden	src/parcelflow/app.py src/parcelflow/auth.py	REQ-AUTH-1	ST, SIT, UAT, REGRESSION	high	security-pack	-	2
17/31	TC-017	Quote error responses never leak secret references	automated	An invalid POST /api/quotes response body never contains the literal string 'secret://'.	python -m unittest tests.test_api.QuoteEndpointTests.test_error_response_never_leaks_secret_reference	src/parcelflow/app.py	REQ-SEC-1	ST, SIT, UAT, REGRESSION	high	security-pack	-	2

Rec	Test ID	Title	Type	Description	Selector / Manual steps	Components	Reqs	Cycles	Crit.	Packs	Deps	Dur (s)
18/31	TC-018	Booking confirms successfully end-to-end over real HTTP	automated	POST /api/bookings for a valid quote with a forced success outcome returns 201 with status=CONFIRMED and attempts=1.	python -m unittest tests.test_api.BookingEndpointTests.test_booking_success_path	src/parcelflow/app.py src/parcelflow/gateway.py src/parcelflow/db.py	REQ-BOOK-1	ST, SIT, REGRESSION	high	-	TC-009, TC-012	2
19/31	TC-019	Booking falls back to queued state over real HTTP	automated	POST /api/bookings with simulated outcomes [timeout, timeout] and fallback_enabled=True returns 201 with status=CONFIRMED_FALLBACK.	python -m unittest tests.test_api.BookingEndpointTests.test_booking_falls_back_when_retries_exhausted	src/parcelflow/app.py src/parcelflow/gateway.py	REQ-BOOK-2	ST, SIT, REGRESSION	high	-	TC-011	2
20/31	TC-020	Booking against a nonexistent quote returns 404	automated	POST /api/bookings with a quote_id that was never created returns HTTP 404 and does not call the gateway.	python -m unittest tests.test_api.BookingEndpointTests.test_booking_for_missing_quote_returns_404	src/parcelflow/app.py src/parcelflow/db.py	REQ-BOOK-1	ST, SIT, REGRESSION	medium	-	TC-012	2
21/31	TC-021	Static UI file serving rejects path traversal	automated	GET /ui/../../pyproject.toml (and similar traversal attempts) return 404, never the file outside ui/.	python -m unittest tests.test_api.StaticUiTests.test_path_traversal_outside_ui_dir_is_rejected	src/parcelflow/app.py ui/	-	ST, SIT, UAT, REGRESSION	medium	security-pack	-	2
22/31	TC-022	Storefront consumer adapter completes quote-then-book flow	automated	StorefrontClient.request_quote followed by create_booking against a real running server returns a CONFIRMED booking.	python -m unittest tests.test_consumers.ConsumerAdapterTests.test_storefront_client_requests_quote_and_books_it	consumers/storefront_client.py	REQ-QUOTE-1, REQ-BOOK-1	ST, SIT, REGRESSION	high	-	TC-015, TC-018	3
23/31	TC-023	Operations consumer adapter reads booking status	automated	OperationsClient.get_booking returns the same booking_id and status as were created via the storefront flow.	python -m unittest tests.test_consumers.ConsumerAdapterTests.test_operations_client_reads_booking_status	consumers/operations_client.py	REQ-AUTH-1, REQ-BOOK-1	ST, SIT, REGRESSION	medium	-	TC-022	3
24/31	TC-024	Quote API response conforms to the documented OpenAPI schema	automated	A real POST /api/quotes response validates against components.schemas.QuoteResponseV1 in openapi/openapi.json using the stdlib-only schema-subset validator.	python -m unittest tests.test_contract.RuntimeContractConformanceTests.test_quote_response_matches_documented_schema	src/parcelflow/contract_check.py openapi/openapi.json src/parcelflow/app.py	REQ-QUOTE-1	ST, SIT, REGRESSION	high	-	-	2
25/31	TC-025	Booking API response conforms to the documented OpenAPI schema	automated	A real POST /api/bookings response validates against components.schemas.BookingResponse in openapi/openapi.json.	python -m unittest tests.test_contract.RuntimeContractConformanceTests.test_booking_response_matches_documented_schema	src/parcelflow/contract_check.py openapi/openapi.json src/parcelflow/app.py	REQ-BOOK-1	ST, SIT, REGRESSION	high	-	-	2
26/31	TC-026	SIT runtime snapshot agrees with SIT repository config	automated	Compares the synthetic captured snapshot snapshots/sit-runtime-snapshot.json against config/sit.json for gateway base URL, timeout, retry count, and the express feature flag.	python -m unittest tests.test_runtime_snapshot.RuntimeSnapshotConsistencyTests.test_sit_snapshot_matches_sit_config	snapshots/sit-runtime-snapshot.json config/sit.json src/parcelflow/config.py	REQ-CONFIG-1	SIT, REGRESSION	medium	-	-	1

Rec	Test ID	Title	Type	Description	Selector / Manual steps	Components	Reqs	Cycles	Crit.	Packs	Deps	Dur (s)
27/31	TC-027	UI escapeHtml prevents markup injection	automated	escapeHtml('&') returns '&'. Node built-in test runner, not a browser execution.	node --test ui/assets/validation.test.js (test: "escapeHtml escapes angle brackets and ampersands")	ui/assets/validation.js	-	ST, REGRESSION	low	-	-	1
28/31	TC-028	UI quote form validation rejects non-numeric weight	automated	validateQuoteForm({weightG: 'abc', ...}) returns valid=false with at least one error. Node built-in test runner, not a browser execution.	node --test ui/assets/validation.test.js (test: "validateQuoteForm rejects non-numeric weight")	ui/assets/validation.js	REQ-QUOTE-1	ST, REGRESSION	low	-	-	1

Rec	Test ID	Title	Type	Description	Selector / Manual steps	Components	Reqs	Cycles	Crit.	Packs	Deps	Dur (s)
29/31	TC-029	Manual: run the Postman collection against a local instance	manual	Start the demo server locally, import openapi/postman_collection.json into Postman (or an equivalent HTTP client), set baseUrl and role, and manually execute each request, confirming 2xx responses and plausible field values.	Manual steps: 1. Start the demo server locally with: python -c "from parcelflow import db; from parcelflow.app import create_server; from parcelflow.config import load_environment, default_config_dir; import json; cfg = load_environment('ST', config_dir=default_config_dir()); conn = db.connect('parcelflow-demo.db'); db.init_schema(conn); rate_card = json.load(open('reference-data/rate_card.json')); server = create_server(cfg, rate_card, conn, port=8101, ui_dir='ui'); print('Serving on http://127.0.0.1:8101'); server.serve_forever()" (see README.md "Run locally" section for the same snippet). 2. Open openapi/postman_collection.json in Postman. 3. Set the baseUrl collection variable to the running server's address. 4. Execute 'Health check' and confirm status 200. 5. Execute 'Create quote (domestic standard)' and confirm status 201 and an integer amount_cents. 6. Copy the returned quote_id into 'Create booking' and execute it; confirm status 201. 7. Copy the returned booking_id into 'Get booking by id' and execute it; confirm the status matches.	openapi/postman_collection.json	REQ-QUOTE-1, REQ-BOOK-1	UAT	medium	-	-	600

Rec	Test ID	Title	Type	Description	Selector / Manual steps	Components	Reqs	Cycles	Crit.	Packs	Deps	Dur (s)
30/31	TC-030	Manual: UI role visibility and accessibility walkthrough	manual	Manually open ui/index.html against a running server, switch the role selector, and visually confirm labels, focus order, and ARIA live regions behave sensibly. This is a manual/static check, not an automated browser test.	Manual steps: 1. Start the demo server on a known port. 2. Serve or open ui/index.html pointed at that server. 3. Tab through the form; confirm every input has a visible label and logical tab order. 4. Switch the role selector between customer/ops/admin and submit a quote for each; confirm the UI does not crash and errors render as text (not raw HTML). 5. Confirm the #quote-errors and #quote-result regions have aria-live='polite' and update sensibly.	ui/index.html ui/assets/app.js	-	UAT	low	-	-	600
31/31	TC-031	Environment-driven behavior genuinely follows each environment's config file	automated	For every known environment, loads the real config/<env>.json and asserts EXPRESS availability, quote role restrictions, and gateway retry-attempt bounds all genuinely follow that file's values -- not a hardcoded expectation.	python -m unittest tests.test_environment_behavior.EnvironmentDrivenBehaviorTests	config/st.json config/sit.json config/uat.json src/parcelflow/app.py src/parcelflow/config.py	REQ-QUOTE-3, REQ-AUTH-1, REQ-BOOK-1	ST, SIT, UAT, REGRESSION	high	-	TC-004, TC-008, TC-009	5

END OF CATALOGUE - 31 test case records.